Streetscape Analysis Using AI: Mapping Crosswalk and Pedestrian Infrastructure from Street View and Aerial Imagery

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1. Introduction

Urban planning has long conceptualized walkability in terms of network connectivity, land use, and density, often assuming that intersections are uniformly crossable. This assumption overlooks a critical dimension of pedestrian mobility: the actual feasibility and quality of street crossings at the neighborhood scale. In practice, missing crosswalks, faded markings, and inadequate traffic control create breaks in the pedestrian network that reduce both accessibility and safety. Yet these features remain largely absent from conventional planning datasets because they are difficult to measure at scale. Engineering-grade methods such as drone or LiDAR surveys can produce highly accurate results, but they are costly and rarely feasible for routine, citywide assessments. Most planning agencies lack the budget, staffing, or technical capacity to conduct regular crosswalk audits. In Philadelphia, for example, the city does not maintain a dedicated land-survey engineering capacity for everyday pedestrian infrastructure management; such expertise is typically mobilized only for major construction projects or when required by state or federal regulation. As a result, routine decisions about signalization and crosswalk repainting often rely on outdated historical survey records rather than current conditions.

This study addresses this gap by asking: *How can we systematically measure crosswalk-based connectivity and crossing conditions at scale using street-level and aerial imagery?* We develop a computer vision pipeline that integrates a U-Net segmentation model with a fine-tuned YOLO detection framework to identify crosswalks, stop signs, and traffic signals. This allows us to construct a spatial dataset of crossing infrastructure, including crosswalk locations, approach-level crossing distances, and intersection control types.

Our contribution is twofold. First, we provide a scalable method that transforms visual street features into structured planning data, enabling new forms of measurement for pedestrian infrastructure. Second, we provide a complete, easy-to-implement analytical framework for cities to easily audit their crossing network. While our analysis is primarily descriptive, it represents a first step toward integrating micro-scale infrastructure into planning metrics and opens the door for future work linking crossing conditions to travel behavior and safety outcomes.

The remainder of this report proceeds as follows. Section 2 motivates the research question and situates it in relation to the limitations of existing pedestrian-infrastructure datasets. Section 3 reviews prior work on walkability measurement, street-view computer vision, and aerial-imagery segmentation. Section 4 presents the detection and segmentation results across major Philadelphia neighborhoods. Section 5 discusses the planning implications, and Section 6 outlines the limitations of the workflow.

2. Why This Matters

Walkability is the pedestrian-facing counterpart to road-network accessibility, and widely used instruments such as Walk Score reduce it to the distance from a location to nearby amenities plus coarse network metrics like block length and intersection density. These measures implicitly treat every intersection as equally crossable. In reality, whether a pedestrian can cross safely, legally, and comfortably depends on micro-scale features that Walk Score and most municipal GIS layers do not capture: whether a crosswalk exists, whether its markings are legible, how wide the crossing is, and whether the approach is controlled by a stop sign, traffic signal, or nothing at all.

The absence of this information is not just an academic gap. Crossing distance is a known correlate of pedestrian exposure and injury risk, particularly for older adults and children who need more time to clear the roadway. Faded or missing markings reduce visibility to drivers and erode yielding behavior. Yet in Philadelphia, as in most U.S. cities, there is no routinely maintained citywide dataset of curb-to-curb crossing widths, marking conditions, or intersection-level traffic control. The information that does exist comes from two incompatible sources: infrequently updated municipal GIS layers and expensive engineering surveys commissioned only for specific capital projects.

The research question this study takes up is therefore practical as much as conceptual: **can computer vision applied to freely available aerial and street-view imagery produce intersection-level pedestrian infrastructure data that is detailed enough to support planning decisions, yet cheap and repeatable enough to cover an entire city?** If yes, it offers a screening layer that sits between outdated GIS and costly field survey, a middle tier that OTIS and similar agencies currently lack. This matters for Vision Zero prioritization, ADA transition planning, and any initiative that needs to know *where* pedestrian infrastructure is failing before committing engineering resources to measure it.

3. Literature Review

Three strands of prior work converge on this project.

Walkability measurement. Early walkability indices (Frank et al., 2006; Glazier et al., 2014) combined intersection density, residential density, and land use mix into composite scores, and commercial products such as Walk Score operationalized the distance-to-amenities view that dominates in practice. More recent literature has argued that these network-and-density formulations overlook the experiential quality of the pedestrian environment, sidewalk continuity, crossing treatments, pedestrian-scale detail, and have pushed toward audit-based instruments such as PEDS and MAPS (Clifton et al., 2007; Cain et al., 2014). Audits capture what network measures cannot, but they are labor-intensive and rarely deployed at a citywide scale. Our work is a response to exactly this tension: the audit literature tells us *what* to measure; the computer vision literature offers the tools to do so at scale.

Street-view computer vision for the urban environment. Google Street View and equivalent panoramic imagery have become a standard substrate for urban analytics since the Streetscore work of Naik et al. (2014) and the Treepedia project at MIT (Li et al., 2015). Street-view-based studies have quantified greenery, sky-view factor, perceived safety, and building facades. More recently, object detection models, including the YOLO family, have been applied to identify traffic signs, signals, and safety hardware (Campbell et al., 2019; Biljecki & Ito, 2021). Our detection component builds directly on this trajectory, but we focus on traffic-control features specifically because they mediate *whether* a crossing exists as a functional pedestrian facility, not just a legal one.

Semantic segmentation of aerial imagery. Convolutional networks, and U-Net in particular (Ronneberger et al., 2015), have become the default tool for extracting linear and polygonal features from remote-sensing imagery like roads, buildings, parcels, and road-surface markings. A small but growing body of work has applied segmentation to crosswalk extraction from satellite or drone imagery (Berriel et al., 2017; Hosseini et al., 2022), typically reporting strong pixel-level accuracy on held-out tiles but limited generalization across cities or camera geometries. While these studies often achieve high pixel-level accuracy on held-out tiles, their performance is less stable across cities and imaging geometries. Our segmentation component builds on the U-Net framework and applies it to super-high-resolution aerial imagery.

The novelty of this project is not in any single model but in the combination: pairing aerial segmentation (which gives geometry, where the crosswalk is and how wide the approach is) with street-view detection (which gives context, what kind of control the crossing has), and binding both to the street-network intersection graph so the outputs are usable by planners.

4. Results

We report results in two parts, corresponding to the pipeline's two modeling components. YOLO-based traffic-control detection was run across 14 Philadelphia neighborhoods and districts; U-Net crosswalk segmentation was trained on a University City pilot and deployed to 3 larger aerial mosaics covering most of the study area.

4.1 Traffic-Control Detection (YOLO)

The YOLOv8-medium detector was applied to Google Street View imagery sampled at four compass-aligned arms per intersection, with a detection confidence threshold of 0.5. Before deployment, we benchmarked the stock COCO-trained model on the COCO validation set to establish a class-specific reference point. The relevant classes behave differently: stop signs are detected with high precision and recall, while traffic lights are harder, with a recall below 0.5.

Class	Precision	Recall	F1	AP@50	mAP@50–95
traffic light	0.725	0.483	0.580	0.569	0.322
stop sign	0.818	0.720	0.766	0.795	0.715

Stop-sign detection is substantially stronger than traffic-light detection at every level. Stop signs are large, planar, high-contrast objects mounted near eye level. The YOLO model handles them well (F1 0.77, AP@50 0.79). Traffic lights are a known YOLO weak point: small in the frame, mounted high, and frequently occluded by tree canopies or building overhangs. A precision of 0.73 is acceptable, but a recall of only 0.48 means nearly half of the real traffic lights in the COCO validation set are missed. These benchmark numbers set expectations for the deployment results that follow: stop-sign counts should be interpreted as close to the true signal, while traffic-light counts are a **conservative lower bound** attributable to the detector rather than to the Philadelphia streetscape.

Across the fourteen study areas, we processed **9,950 street-view images** and produced **3,745 total detections**, including **2,901 traffic lights** across 1,481 images and **844 stop signs** across 780 images. Aggregated citywide, the traffic-light detection rate is **14.9%** of sampled images (mean confidence 0.677) and the stop-sign rate is **7.8%** (mean confidence 0.712, slightly higher than traffic lights). At the image level, stop signs average ~1.0 instance per positive image, whereas traffic lights average 1.8–2.4 instances, consistent with signalized intersections typically carrying a *cluster* of signal heads (one per approach lane), while stop-controlled approaches have a single sign.

Average detection confidence was stable across neighborhoods (0.65–0.71 for traffic lights, 0.61–0.80 for stop signs), suggesting that between-neighborhood differences reflect real variation in installed infrastructure rather than systematic detector drift.

4.2 Crosswalk Segmentation (U-Net)

The crosswalk segmentation model is a U-Net with a ResNet-34 encoder, trained on 256×256 patches derived from PASDA 2024 aerial imagery over a University City pilot scene. We manually labeled 202 crosswalk polygons, rasterized them into per-pixel binary masks, generated 934 image/mask patch pairs, and partitioned them into 653 training, 140 validation, and 141 test patches using stratified sampling by crosswalk coverage.

Training ran for 25 epochs with binary cross-entropy and Dice loss; the best validation Dice score of **0.919** was achieved at epoch 17 and used as the deployed checkpoint. A post-hoc threshold sweep on the validation split identified **τ = 0.40** as the F1-optimal decision threshold, with the following pixel-level test metrics:

Threshold	Precision	Recall	F1 (Dice)	IoU
0.40	0.924	0.955	**0.939**	0.886
0.45	0.928	0.951	0.939	0.886
0.50	0.932	0.946	0.939	0.885

On a held-out twelve-patch visualization sample, mean Dice was 0.816 and mean IoU was 0.763. The gap between patch-level mean and aggregated Dice reflects the fact that most error is concentrated in a small number of hard patches — partially occluded crossings, novel marking styles (continental vs. ladder), and patches containing only a sliver of crosswalk.

The deployed model was then applied to three large mosaics covering the middle, southern, and upper-middle portions of the study area. Sliding-window inference at a 256-pixel patch size and a 128-pixel stride produced pixel-level probability rasters, which were thresholded at 0.45, cleaned by removing connected components smaller than 200 pixels, morphologically closed, and vectorized to polygons.

Across the three deployment scenes the model produced **3,610 crosswalk polygons** — a first-pass citywide(-scale) crosswalk inventory where none previously existed at this resolution. These are written as `<scene>_crosswalks.gpkg` / `.shp`, with per-polygon area in square feet, suitable for direct use in ArcGIS or QGIS.

5. Discussion

The results support the core claim of the project: a combined aerial-segmentation plus street-view-detection pipeline can produce intersection-level pedestrian infrastructure data at a spatial grain that existing municipal GIS layers do not reach, at a cost and cadence that field engineering surveys cannot match. Three findings are worth drawing out.

Detection patterns recover known streetscape structure. The commercial-core vs. rowhouse-neighborhood inversion documented in Section 4.1 — signalized control clustering in University City, Logan Square, Center City, and Rittenhouse; stop-sign control clustering in Kensington, West Kensington, Fishtown, and Point Breeze — is not something we imposed. It emerges directly from the detector output and matches what anyone who has walked these neighborhoods already knows. That the pipeline reproduces this pattern *asymmetrically* (with the signal side attenuated by the detector's known recall weakness, yet still clearly visible) gives some confidence that the stop-sign side, where the model is strong, is close to a census. For planners, the practical implication is that the workflow can be used to map *where a given control type dominates* — useful for pedestrian-signal retiming studies, stop-sign compliance audits, or identifying uncontrolled intersections for midblock-crossing studies — with the caveat that signal counts should be treated as a lower bound.

Crosswalk segmentation generalizes within a single imagery program. The pixel-level F1 of 0.94 on the University City test set is strong, and the fact that applying the same model to three much larger mosaics (covering roughly 10× the training area) yielded coherent polygon inventories — 3,610 polygons with no manual correction — indicates that within a single aerial imagery program (PASDA 2024, ~15 cm/pixel, consistent lighting and capture season), transfer is practical. This is the right unit of transfer to reason about: annual PASDA captures mean that the same pipeline can, in principle, be re-run each year to track marking degradation and new installations without retraining.

For OTIS and similar agencies, the immediate use case is screening. Rather than asking "where should we send a survey crew?", an analyst can filter the crossings layer for long crossing distances with low marking coverage and no signal control — a plausible proxy for high-risk uncontrolled crossings — and prioritize field inspection accordingly. The workflow does not replace engineering measurement; it tells you where engineering measurement is most likely to be worth the expense.

6. Limitations

Several limitations qualify these results.

No independent ground-truth evaluation at deployment scale. The 0.94 Dice reported in Section 4.2 is a held-out *patch-level* metric within the University City pilot. We have not produced an independent field-validated sample for the larger deployment mosaics, so the pixel-level performance on those scenes is an extrapolation. Similarly, the YOLO detection numbers are reported on the COCO val2017 set, not on a Philadelphia-specific, hand-labeled evaluation set. Both models would benefit from a small, location-stratified validation audit before any planning decisions are made regarding the outputs.

Detection is biased by imagery availability. Google Street View coverage is uneven in both space and time — some arms have 2019 imagery, others 2023 — and panorama capture dates are mixed across our fourteen neighborhoods. A stop sign installed in 2022 will appear or not, depending on which pano Google served. For a monitoring application this is a hard constraint: the pipeline is only as current as the underlying imagery.

Traffic-light recall is substantially lower than stop-sign recall. The COCO benchmark shows traffic-light recall of 0.48 compared to 0.72 for stop signs, and field experience suggests traffic-light detection is degraded further by viewing geometry (signals mounted overhead or far from the approach) and by confusion with other illuminated street furniture. Deployment counts for traffic lights should be treated as a lower bound rather than a census.

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